Lesson 4: Direct Hit

TIC-80

VIDEO GAME CREATION

This week: detecting when things touch. Your job is to add one enemy to the screen and make your bullet destroy it on contact - the first piece of your capstone's actual combat loop.

Open your own [YourName]Game project from Lesson 3 - don't start a new one. Everything this week builds on the player and bullet you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder enemy image (enemy_invader_red.png) and a new sound effect (sfx_hit.wav) - you don't need to find or make your own art today.

Setup

1. Launch TIC-80 and load your cart from Lesson 3.
2. Open the Sprite Editor (F2) and draw a small enemy shape on sprite index 3 (index 1 is your player, index 2 is your bullet).

Level 1 - Minimal (MVP)

A single enemy sits on screen. When your bullet touches it, both the bullet and the enemy disappear. This is a complete finish for today - reaching this is enough to move on to next week.

3. Press Esc to reach the code editor.
4. Add enemy_alive=true and fixed ex, ey coordinates near the top of your code, above the TIC() function - this is the enemy's one and only position for this lesson.
5. Inside your bullets loop (from Lesson 3), before the existing off-screen check, add a simple rectangle-overlap check against ex/ey - if it overlaps and the enemy is still alive, set enemy_alive=false and remove that bullet from the table.
6. At the bottom of TIC(), draw the enemy only if enemy_alive is still true.
7. Press Esc, type run, press Enter, then shoot the enemy to test - both the bullet and the enemy should disappear on contact.

```
enemy_alive=true
ex,ey=60,10
bullets={}

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
```

```
  table.insert(bullets,{bx=x,by=y})
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  if enemy_alive and b.bx<ex+8 and b.bx+8>ex and b.by<ey+8 and b.by+8>ey
then
   enemy_alive=false
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 if enemy_alive then spr(3,ex,ey,0) end
 spr(1,x,y,0)
end
```

Level 2 - Complete (Alpha/Beta)

The enemy is destroyed cleanly and exactly once, even if you fire several bullets in a row - no errors, no lingering references, and a bullet that misses still behaves exactly like it did last week.

- Fire several bullets in a row, including some that miss - the loop walks every bullet in the table each frame, so a hit only ever removes the one bullet whose position actually overlaps the enemy, and every other bullet keeps moving and gets cleaned up off-screen exactly like it did in Lesson 3.
- The enemy_alive check and the overlap check together mean that once enemy_alive is false, that branch can never run again for any later bullet - so the enemy can only ever be "destroyed" once, even if two bullets happen to overlap it on the very same frame.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: design a short impact sound in the SFX Editor (F4) in an empty slot (e.g. slot 2, since slot 1 is likely your shoot sound), then call sfx(2) right where enemy_alive is set to false.
- Visual: draw enemy sprite index 3 as visually distinct from the player and bullet (different shape or color) in the Sprite Editor - this alone satisfies most of "its own look" without any extra code.
- Structure: keep the enemy's own state (enemy_alive, ex, ey) declared once at the top of your code, and only ever change enemy_alive from inside the collision check shown above - resist checking or setting it from more than one place, even though everything still technically lives inside the single TIC() function.

If Something's Wrong

- Enemy vanishes but no bullet was actually near it: double check the overlap math is comparing bullet coordinates (b.bx, b.by) against the enemy's (ex, ey), not against the player's (x, y).
- Bullets stop appearing once the enemy is destroyed: confirm the enemy_alive check only guards the collision branch and the drawing line at the bottom, not the whole bullets loop.

- "attempt to compare nil with number" error: confirm ex,ey=60,10 (or your chosen coordinates) is declared before TIC() runs, same as bullets={}.

Want More?

Student Resource List > TIC-80 section, or tic80.com